SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Concept Mapping

Course Code:	CSA5802	**Course Title:**	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 3
Weightage:	40% of CIA	**Total Marks:**	100
CO1 Statement:	*Concept Mapping, Reinforcement Learning Framework Visualization and GitHub Submission.*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Instructions to Students

- One submission per team (3–5 students).
- Ensure diagrams are original and professionally formatted.
- Include all editable source files.
- Submit the GitHub repository linkand pdf through the classrome before the deadline.
- Late submissions may be subject to the course policy.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Creative Things and concept grasping	Q1	Q1 –100
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Mandatory Concepts

- Planning
- Requirements
- Threat Modeling
- Secure Design
- Secure Coding
- Git
- GitHub
- Branch
- Pull Request
- Continuous Integration (CI)
- Software Composition Analysis (SCA)
- Secret Scanning
- Docker
- Container Scanning
- Continuous Deployment (CD)
- Kubernetes

- Deployment
- Monitoring
- Logging
- SIEM
- Feedback Loop
-

Reinforcement Learning Framework Visualization

Create a feedback-loop diagram that demonstrates continuous improvement in the DevSecOps pipeline. The visualization should show how security findings from SAST, SCA, container scanning, runtime monitoring, logging, and incident analysis are fed back to developers for remediation, resulting in an iterative cycle of secure software development. Note: This activity does not require implementing a machine learning algorithm.

Suggested Repository Structure

```
DevSecOps-AT1/
│
├── README.md
├── Concept_Map.pdf
├── RL_Framework.pdf
├── Documentation.pdf
├── Source/
│   ├── concept_map.drawio (or equivalent)
│   └── rl_framework.drawio (or equivalent)
└── Images/
```

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Rubrics:

Criteria	Weightage	Awarded Marks	Total Marks (100)
Understanding of Concepts	25%		
Experimental Design	30%		
Use of Tools	20%		
Analysis of Results	15%		
Documentation	10%		
Total per Question	100 Marks		

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100

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________